Selektywność (Selectivity)

Jak bardzo indeks zawęża wyszukiwanie przygotowując rezultat.

Im wyższa wartość selektywności tym lepiej (max 1)

distinct_codes	rows_per_code	selectivity
10	1,000,000	0.00001

```
SELECT count(DISTINCT product_code) distinct_codes,
       count(*) rows_per_code,
       count(DISTINCT product_code)::float/count(*) AS selectivity
  FROM products;
```

Selektywność (Selectivity)

```
SELECT product_code, count(*)::float/1000000 AS values_distribution
  FROM products
 GROUP BY product_code
```

product_code	values_distribution
PRD1	0.099985
PRD10	0.099986
PRD2	0.099576
PRD3	0.100206
PRD4	0.100257
PRD5	0.099956
PRD6	0.100038
PRD7	0.099669
PRD8	0.100046
PRD9	0.100281

Selektywność (Selectivity) – jak zwiększyć?

```
CREATE INDEX idx_prd_composite_prd_code_name ON products (product_code, product_name);

SELECT count(DISTINCT product_name||'_'||product_code) distinct_codes,
       count(*) rows_per_code,
       count(DISTINCT product_name||'_'||product_code)::float/count(*) AS selectivity
  FROM products;
```

distinct_codes	rows_per_code	selectivity
99,999	1,000,000	0.099999